M.S. NAZ HIGH SCHOOL

Excellence in Matriculation & Academic Leadership | Est. 1989 | BISE Gujranwala Code: 112199

Single National Curriculum (SNC) & PECTAA Standards Aligned - Session 2026-2027

CLASS 9 (9TH MATRIC) - CHEMISTRY

Official Syllabus, Board Paper Scheme & High-Yield Examination Revision Notes

Group: Science Group | Total Marks: 75 Marks (Theory: 60 | Practical: 15) | Time: 2 Hours 45 Minutes

SECTION 1: BISE GUJRANWALA EXAMINATION BLUEPRINT & WEIGHTAGE

- **Section A: Objective (MCQs):**
  12 Compulsory MCQs (1 Mark each = 12 Marks).
- **Section B: Short Questions:**
  Attempt 15 out of 24 Short Questions (3 sub-sections: Q2, Q3, Q4: 5/8 each = 30 Marks).
- **Section C: Long Questions:**
  Attempt 2 out of 3 Long Questions (Q5, Q6, Q7: 9 Marks each = 18 Marks).
- **SLO Cognitive Weightage:**
  Knowledge: 45% | Understanding: 40% | Chemical Equation Balancing & Application: 15%.

SECTION 2: CHAPTER-WISE HIGH-YIELD REVISION NOTES & CORE SLOs

Chapter 1: Fundamentals of Chemistry:

Branches of Chemistry, Elements, Compounds vs Mixtures, Atomic Number (Z) vs Mass Number (A), Relative Atomic Mass and amu, Chemical Formula writing, Empirical vs Molecular formula, Mole and Avogadro number (NA = 6.02 x 10^23), Mole calculations.

Chapter 2: Structure of Atoms:

Rutherford Atomic Model & its defects, Bohr Atomic Theory and Postulates, Electronic Configuration of first 18 elements (s, p, d, f subshells), Isotopes of Hydrogen, Carbon, Chlorine, Uranium and their industrial/medical uses.

Chapter 3: Periodic Table & Periodicity:

Modern Periodic Law, Groups (1 to 18) and Periods (1 to 7), s, p, d, f blocks, Periodic Trends: Atomic radius, Ionic radius, Ionization energy, Electron affinity, Electronegativity (variation along periods and down groups).

Chapter 4: Structure of Molecules:

Octet and Duplet rule, Why atoms react, Chemical Bonds: Ionic Bond (NaCl), Covalent Bond (Single, Double, Triple), Dative/Coordinate Covalent Bond (NH4+, H3O+), Metallic Bond (electron sea model), Intermolecular Forces (Dipole-Dipole, Hydrogen Bonding in water).

Chapter 5: Physical States of Matter:

Kinetic Molecular Theory of gases, Boyle Law (P1V1 = P2V2), Charles Law (V1/T1 = V2/T2, Kelvin scale), Absolute Zero (-273.15 deg C), Diffusion, Effusion, Liquid state (evaporation, vapor pressure, boiling point), Solid state (amorphous vs crystalline, allotropy of Sulfur, Carbon, Tin).

Chapter 6: Solutions:

Solute vs Solvent, Types of solutions (gas, liquid, solid), Saturated, Unsaturated, and Supersaturated solutions, Concentration Units: Percentage (% m/m, % m/v, % v/m, % v/v), Molarity (M = moles of solute / dm^3 of solution), Dilution of solution (M1V1 = M2V2), Solubility and "Like dissolves like", Colloids and Suspensions (Tyndall Effect).

Chapter 7: Electrochemistry:

Oxidation and Reduction in terms of electron loss/gain and oxidation state, Rules for assigning oxidation number, Oxidizing vs Reducing agents, Electrolytic Cell vs Galvanic (Voltaic) Cell, Nelson cell for NaOH manufacture, Down cell for Na, Rusting of iron and prevention (galvanizing, tin plating, sacrificial protection).

Chapter 8: Chemical Reactivity:

Metals and Non-metals properties, Electropositive character of metals and reactivity series, Alkali vs Alkaline earth metals, Inertness of noble metals (Gold, Platinum), Non-metals (Halogens reactivity trend F2 > Cl2 > Br2 > I2).

SECTION 3: MSNS SENIOR FACULTY BOARD EXAMINATION STRATEGY

*** Write balanced chemical equations with state symbols (s, l, g, aq) for every chemical reaction.**

*** In electronic configuration questions, always specify principal energy levels (K, L, M) and subshells (1s, 2s, 2p).**

*** For Molarity numericals, always convert volume from cm^3 to dm^3 by dividing by 1000.**